

20-Day Study Plan — 2 to 21 Oct 2026

Goal: steady progress on the [SDE Interview & Prep Roadmap](#) and [CMU 15-445/645 Fall 2025](#) (all 24 lectures, homeworks 1–4, projects 0–2), plus a small daily career slot.

Budget: 3h Sun–Thu after work, 7h on Fri/Sat = **84h**; every day below adds up exactly.

Out of scope: repo domains 1 (Data Structures), 2 (Algorithms) and 9 (Problem-solving and Coding). Contest-Practice stays.

Milestones

Day 2	Sat 3 Oct	P0 C++ Primer passes on Gradescope
Day 5	Tue 6 Oct	HW1 SQL + HW2 Storage self-graded; three CV versions done
Day 7	Thu 8 Oct	IELTS booked
Day 11	Mon 12 Oct	P1 Buffer Pool Manager passes on Gradescope
Day 13	Wed 14 Oct	HW3 Indexes self-graded
Day 19	Tue 20 Oct	HW4 self-graded; DAAD motivation letter final
Day 20	Wed 21 Oct	P2 B+Tree passes on Gradescope; all 24 lectures watched

Time by track

CMU Project	29h
CMU Lecture	24h
Career	9h
Repo	8h
CMU Homework	8h
Contest Practice	4.5h
Review	1.5h
Total	84h

Daily routine

- Weekday (3h, after work):** 1 lecture at 1.5× with notes (1h) → project or homework block (1–1.5h) → repo domain (30 min) or career slot (30 min).
- Fri / Sat (7h):** 2 lectures (2h) → deep project block (3–4h) → virtual contest from the Contest-Practice PDFs or a repo domain (1–1.5h) → career slot (30 min).
- If you slip:** move the task to the next free slot. Never skip a project checkpoint. Day 20 is the buffer.

Key links

- CMU 15-445 Fall 2025: [schedule](#) · [assignments](#) · [FAQ](#) · [lecture playlist](#)
- Gradescope for non-CMU students:** entry code **5R4XPZ**, school "Carnegie Mellon University".
- Projects: [P0 C++ Primer](#) · [P1 Buffer Pool Manager](#) · [P2 B+Tree Index](#) · [P3 Query Execution](#) · [P4 Concurrency Control](#)
- Repo: [README checklist](#) · [Contest practice PDFs](#) · [Application tracker](#) · [Strategy](#)
- Keep your BusTub copy **private** (course academic-integrity policy).

After day 20

- ☐ Days 21–27: CMU P3 Query Execution (Tasks 1–2) + HW5 Transactions — [link](#)
- ☐ Days 21–27: Submit DAAD before 30 Oct — [link](#)
- ☐ Days 28–34: CMU P3 finish + P4 Concurrency Control start — [link](#)
- ☐ Days 28–34: Repo: System Design domain — scalability, load balancing, caching, DB design — [link](#)
- ☐ Days 35–40: CMU P4 finish + HW6 Recovery + L25 final review — [link](#)
- ☐ Days 35–40: Repo: Programming Languages & Concepts (OS + Networking are now done in days 1–13) — [link](#)
- ☐ November: ETH Zurich ESOP application window (1–30 Nov) — [link](#)
- ☐ Days 41–50: Repo: System Architecture — [link](#)
- ☐ Days 41–50: Contest practice — 2 virtual contests a week (Contest-Practice PDFs) — [link](#)
- ☐ Days 51–60: Mock interviews — 1 system design + 1 behavioral (STAR) per week — [link](#)
- ☐ Days 51–60: Repo: final checklist sweep (domains 3–8 and 10) — [link](#)
- ☐ Days 51–60: Re-run the application tracker; follow up on all applications — [link](#)

Live version with checkboxes and calendar: the "20-Day Plan — 2 to 21 Oct 2026" page in Notion. Tick ☐ as you go.

Day 1 · Fri 2 Oct 2026			7h · weekend
☐	CMU Lecture	L01 — Relational Model & Algebra Reading: Chapters 1-2 Tick: Relational databases (SQL)	1h Video · Slides · Notes
☐	CMU Lecture	L02 — Modern SQL Reading: Chapters 3-5 Tick: DQL, DML, DDL, Joins, Views	1h Video · Slides · Notes
☐	CMU Project	P0 C++ Primer — Task #1 (first pass)	2.5h Project spec
☐	CMU Project	Setup: clone BusTub into a private repo, build (CMake + clang), gdb ready; join non-CMU Gradescope (code 5R4XPZ, school: Carnegie Mellon University) Follow "Creating Your Own Project Repository" + "Setting Up Your Development Environment" on the P0 page	1h Project spec
☐	Repo	OS: Processes + Threads guides Tick: Operating Systems → Processes; Threads	1h Guide · Threads guide
☐	Career	Plan the week in this board; pick which CV version to finish first	0.5h Roadmap file
Day 2 · Sat 3 Oct 2026			7h · weekend
☐	CMU Lecture	L03 — Database Storage I Reading: Chapter 12.1-12.4, 13.2-13.3	1h Video · Slides · Notes
☐	CMU Lecture	L04 — Memory Management Reading: Chapter 13.2-13.5	1h Video · Slides · Notes
☐	CMU Project	P0 — finish, run tests + make check-clang-tidy + memory-leak check, submit to Gradescope (make submit-p0) Milestone: P0 passes on Gradescope	1.5h Project spec
☐	CMU Homework	HW1 SQL (DuckDB) — Q1–Q4 Tick: Databases → DQL / Joins	1.5h Homework
☐	Contest Practice	Contest PDF #1: CF Round 1074 (Div. 4) — virtual, solve A–D, then read hints	1.5h Contest PDFs
☐	Career	20-min career slot: shortlist 5 P1 companies from the tracker	0.5h Roadmap file
Day 3 · Sun 4 Oct 2026			3h · weekday
☐	CMU Lecture	L05 — Database Storage II Reading: Chapter 14.8.1, 24.2	1h Video · Slides · Notes
☐	CMU Homework	HW1 — Q5–Q6, then compare with the official solutions	1h Homework · Solutions
☐	Repo	OS: Thread synchronization (mutexes, semaphores, monitors) Tick: Operating Systems → Thread synchronization mechanisms (Mutexes, Semaphores, Monitors)	0.5h Guide
☐	Career	CV version 1: General SWE (one page)	0.5h Roadmap file
Day 4 · Mon 5 Oct 2026			3h · weekday
☐	CMU Lecture	L06 — Storage Models & Compression Reading: Chapter 11.2, 13.6	1h Video · Slides · Notes
☐	CMU Homework	HW2 Storage — first half	1h Homework
☐	Repo	OS: Scheduling algorithms Tick: Operating Systems → Scheduling algorithms (Fair share, EDF, WFQ)	0.5h Guide
☐	Career	Knight-Hennessy (deadline 6 Oct) — go/no-go (optional stretch); CV version 2: Systems/Rust	0.5h Roadmap file

Day 5 · Tue 6 Oct 2026			3h · weekday
☐	CMU Lecture	L07 — Hash Tables Reading: Chapter 14.5, 24.5	1h Video · Slides · Notes
☐	CMU Homework	HW2 Storage — finish + self-grade with the official solutions	1h Homework · Solutions
☐	Repo	OS: Memory management (page replacement, mmap, buddy allocation) Page replacement (LRU, Clock) is the same idea as the CMU P1 replacer. Tick: Operating Systems → Memory management (Page replacement, Memory-mapped files, Buddy allocation)	0.5h Guide
☐	Career	CV version 3: Embedded	0.5h Roadmap file
Day 6 · Wed 7 Oct 2026			3h · weekday
☐	CMU Project	P1 Buffer Pool Manager — read the spec; work through the ARC algorithm on paper	1h Project spec
☐	Repo	Databases domain: read SQL-DQL, SQL-DML, Joins, Normalization guides; tick what HW1 + L01–L02 covered Tick: Databases → Relational databases (SQL) sub-items	1h Guide
☐	Repo	OS: Deadlock detection and prevention Tick: Operating Systems → Deadlock detection and prevention	0.5h Guide
☐	Career	Send 1 application (P1 list) with the matching CV	0.5h Roadmap file
Day 7 · Thu 8 Oct 2026			3h · weekday
☐	CMU Project	P1 Task #1 — ARC replacer: implementation start	1.5h Project spec
☐	Repo	OS: File systems; tick all finished OS checklist items Tick: Operating Systems → File systems (Journaling, NFS/SMB, Encryption, HDFS/Ceph)	0.5h Guide
☐	Career	Book the IELTS test (needed for master's / Canada; target ~15 Oct)	0.5h Roadmap file
☐	Review	Weekly review #1: tick finished checklist items in the repo README; move unfinished tasks to the next buffer slot	0.5h README checklist
Day 8 · Fri 9 Oct 2026			7h · weekend
☐	CMU Lecture	L08 — Indexes & Filters I Reading: Chapter 14.1-14.4 Tick: Indexing → B-tree, B+ tree	1h Video · Slides · Notes
☐	CMU Lecture	L09 — Indexes & Filters II Reading: Chapter 14.1-14.4, 24.1 Tick: Indexing → Bitmap indexing	1h Video · Slides · Notes
☐	CMU Project	P1 Task #1 — ARC replacer: finish + pass local tests	2h Project spec
☐	CMU Project	P1 Task #2 — Disk Scheduler	1h Project spec
☐	Contest Practice	Contest PDF #2: CF Round 1062 (Div. 4) — virtual, A–D	1.5h Contest PDFs
☐	Career	Send 1 application	0.5h Roadmap file
Day 9 · Sat 10 Oct 2026			7h · weekend
☐	CMU Lecture	L10 — Index Concurrency Control Reading: Chapter 18.10.2	1h Video · Slides · Notes
☐	CMU Lecture	L11 — Sorting & Aggregations Algorithms Reading: Chapter 15.4-15.5	1h Video · Slides · Notes
☐	CMU Project	P1 Task #3 — Buffer Pool Manager + page guards	4h Project spec
☐	Repo	Networking: TCP/IP stack (OSI layers, TCP vs UDP) Tick: Networking → TCP/IP stack (OSI model layers, TCP vs. UDP)	0.5h Guide
☐	Career	Send 1 application	0.5h Roadmap file

Day 10 · Sun 11 Oct 2026			3h · weekday
☐	CMU Lecture	L12 — Joins Algorithms Reading: Chapter 15.4-15.6 Tick: Joins	1h Video · Slides · Notes
☐	CMU Project	P1 — concurrency + debugging (gdb, sanitizers)	1h Project spec
☐	Repo	Networking: DNS (resolution process) Tick: Networking → DNS (Resolution process)	0.5h Guide
☐	Career	Send 1 application	0.5h Roadmap file
Day 11 · Mon 12 Oct 2026			3h · weekday
☐	CMU Lecture	L13 — Query Execution I Reading: Chapter 15.1-15.3, 15.7	1h Video · Slides · Notes
☐	CMU Project	P1 — pass all Gradescope tests + format + leak check, submit Milestone: P1 passes on Gradescope	1.5h Project spec
☐	Career	Send 1 application	0.5h Roadmap file
Day 12 · Tue 13 Oct 2026			3h · weekday
☐	CMU Lecture	L14 — Query Execution II Reading: Chapter 22	1h Video · Slides · Notes
☐	CMU Homework	HW3 Indexes & Filters — first half Tick: Databases → Indexing	1h Homework
☐	Repo	Networking: HTTP (methods, status codes) Tick: Networking → HTTP protocol (Request methods, Status codes)	0.5h Guide
☐	Career	Send 1 application	0.5h Roadmap file
Day 13 · Wed 14 Oct 2026			3h · weekday
☐	CMU Lecture	L15 — Query Planning & Optimization I Reading: Chapter 16 Tick: Database design and optimization	1h Video · Slides · Notes
☐	CMU Homework	HW3 — finish + self-grade	1h Homework · Solutions
☐	Repo	Networking: Routing algorithms; tick all finished Networking items Tick: Networking → Routing algorithms (Shortest path: Dijkstra, Bellman-Ford)	0.5h Guide
☐	Career	DAAD (deadline 30 Oct): document checklist + programme shortlist	0.5h Roadmap file
Day 14 · Thu 15 Oct 2026			3h · weekday
☐	CMU Lecture	L16 — Query Planning & Optimization II Reading: Chapter 16 Tick: Materialized views (partly)	1h Video · Slides · Notes
☐	CMU Project	P2 B+Tree — read the spec + Development Roadmap; Task #1 B+Tree pages	1h Project spec
☐	Career	Send 1 application	0.5h Roadmap file
☐	Review	Weekly review #2: tick checklist items, re-plan slips	0.5h README checklist
Day 15 · Fri 16 Oct 2026			7h · weekend
☐	CMU Lecture	L17 — Concurrency Control Theory Reading: Chapter 18 Tick: Transactions; ACID properties	1h Video · Slides · Notes
☐	CMU Lecture	L18 — Two-Phase Locking Concurrency Control Reading: Chapter 18.1-18.3, 18.9 Tick: Isolation, Isolation levels	1h Video · Slides · Notes
☐	CMU Project	P2 Task #1 finish; Task #2 point search + insertion (with splits)	3h Project spec
☐	Contest Practice	Contest PDF #3: ABC 476 — virtual, A–E	1.5h Contest PDFs
☐	Career	Send 1 application	0.5h Roadmap file

Day 16 · Sat 17 Oct 2026			7h · weekend
☐	CMU Lecture	L19 — Timestamp Ordering Concurrency Control Reading: Chapter 18.5-18.6	1h Video · Slides · Notes
☐	CMU Lecture	L20 — Multi-Version Concurrency Control Reading: Chapter 18.7-18.8 Tick: Isolation levels (MVCC)	1h Video · Slides · Notes
☐	CMU Project	P2 Task #2 — deletion (merge / redistribute); use the tree visualizer	3.5h Project spec
☐	Repo	Version Control domain: Git guide + command reference, Bitbucket Tick: Version control System → Git; Bitbucket	1h Guide · Command reference · Bitbucket
☐	Career	Send 1 application	0.5h Roadmap file
Day 17 · Sun 18 Oct 2026			3h · weekday
☐	CMU Lecture	L21 — Database Logging Reading: Chapter 19.1-19.8 Tick: Atomicity, Durability	1h Video · Slides · Notes
☐	CMU Project	P2 Task #3 — Index Iterator	1.5h Project spec
☐	Career	DAAD motivation letter — first draft	0.5h Roadmap file
Day 18 · Mon 19 Oct 2026			3h · weekday
☐	CMU Lecture	L22 — Database Recovery Reading: Chapter 19.1-19.9 Tick: Durability; Transaction Control	1h Video · Slides · Notes
☐	CMU Project	P2 Task #4 — Concurrency control (latch crabbing)	1.5h Project spec
☐	Repo	System Design: Database Design and Optimization guide (ties to CMU L15–L16) Tick: System Design → Database design and optimization	0.5h Guide
Day 19 · Tue 20 Oct 2026			3h · weekday
☐	CMU Lecture	L23 — Distributed Database Systems I Reading: Chapter 20.4-20.5, 21, 23.1-23.4 Tick: Partitioning; NoSQL databases → Types	1h Video · Slides · Notes
☐	CMU Homework	HW4 Execution & Planning — solve + self-grade	1.5h Homework · Solutions
☐	Career	DAAD motivation letter — final; upload documents	0.5h Roadmap file
Day 20 · Wed 21 Oct 2026			3h · weekday
☐	CMU Lecture	L24 — Distributed Database Systems II Reading: Chapter 20.7, 22.9 Tick: NoSQL databases → Examples	1h Video · Slides · Notes
☐	CMU Project	Buffer: P2 — pass Gradescope + leak check, submit (or catch up on anything that slipped) Milestone: P2 passes on Gradescope	1.5h Project spec
☐	Review	Final review: tick all finished repo checklist items; write the plan for days 21–40	0.5h README checklist